

BADAL SAHANI

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PROFESSIONAL SUMMARY

Full-Stack Developer who built and shipped **Notesify** — a production-grade AI notes app featuring multi-provider LLM orchestration, sub-10ms Redis caching, and a full auth system with refresh token rotation and Google OAuth. Skilled in React, TypeScript, Node.js, Express, and MongoDB, with a track record of diagnosing real performance bottlenecks and shipping measurable improvements.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Java, HTML5, CSS3

Frontend: React.js, TipTap, TanStack React Query, Framer Motion, Tailwind CSS

Backend: Node.js, Express.js, REST API Design, Passport.js, JWT, Redis (Upstash), Nodemailer, bcrypt

Databases: MongoDB, MySQL, PostgreSQL

Tools & Platforms: Git, GitHub, Postman, Vercel, Cloudinary, Google OAuth 2.0

Core Competencies: Data Structures & Algorithms, System Design, OOP

PROJECTS

Notesify — AI-Powered Full-Stack Notes Application

React · TypeScript · TipTap · Node.js · Express · MongoDB · Redis · TanStack Query · Framer Motion · Cloudinary · Passport.js · JWT [Live](#) | [GitHub](#)

- **Multi-provider AI orchestration layer** routing requests dynamically across Groq (Llama 3.3 70B), NVIDIA (Gemma 2 9B), and Google Gemini based on payload type — with cascading fallbacks ensuring near-zero AI downtime
- **Optimistic Concurrency Control (OCC)** using version fields on all core documents — backend returns 409 on version mismatch, frontend performs instant predictive UI updates via TanStack React Query and rolls back only on confirmed conflict
- **Production-grade auth system** with refresh token rotation, automatic reuse detection (replay attack → revoke all sessions), 5-device session capping, Google OAuth account linking, and email verification with 3-tier trust logic
- **Sub-10ms repeat API responses** via Upstash Redis caching with self-healing invalidation — on cache drift (404), auto re-fetches from MongoDB to re-seed Redis without manual intervention
- **SHA-256 content-hashed AI response caching** eliminating redundant LLM API calls; auto-summarizes chat history at 20 messages into a rolling context snapshot to manage token limits
- **Diagnosed and resolved 4 frontend performance bottlenecks** — LCP 3s → 1.7s, INP 0.3 → 0.04, CLS 0.29 → 0.04 — via Map-based O(1) search cache, elimination of redundant auth waterfall (~1.6s scripting time saved), and Framer Motion layoutId removal (~70% CPU reduction on mount)
- **Custom TipTap editor** with 6+ extensions: ghost text generation, drag-and-drop Cloudinary image upload, auto-detected code blocks, resizable tables, task lists, AI bubble menu, and hybrid MongoDB full-text + regex search

EDUCATION

Brainware University, Barasat, Kolkata | Bachelor of Computer Applications (BCA) | CGPA: 7.86 / 10 | Graduated 2025

CERTIFICATIONS

Python Programming — Great Learning (April 2023) · Topics: Data Structures, OOP, Algorithms